

Exact Algebraic Computation of Learning Coefficients for Two-Dimensional Singular Models

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Abstract

Classical information criteria such as the Bayesian Information Criterion (BIC) rely on regularity assumptions that break down for singular models, leading to incorrect model selection in settings such as deep learning. The Widely Applicable Bayesian Information Criterion (WBIC) relies on local learning coefficients λ , which in the analytic case coincides with local Real Log Canonical Thresholds (RLCT) of the Kullback-Leibler divergence of the model, to capture correct marginal likelihood asymptotics. Exact computation of the learning coefficients has been limited to special cases, and only sampling-based estimation methods are generally applicable. We present the first deterministic algorithm that computes local RLCTs exactly for any two-dimensional model whose Kullback-Leibler distance is contact equivalent to a polynomial, derive a bound on its complexity, and demonstrate its effectiveness for a broad class of models, with applications including polynomial neural networks. Beyond providing ground truth to calibrate sampling-based estimators, exact computation reveals algebraic structure in learning coefficients that sampling cannot and out-speeds it in the shallow regime.

1 Introduction

Information criteria (IC): IC are model selection methods in statistics and machine learning [Konishi and Kitagawa, 2008]. They aim to strike a balance between prediction accuracy and model complexity. They are ubiquitous in diverse domains, ranging from statistics [Zhang et al., 2023], especially in the presence of structured learning [Barber and Drton, 2015, De Campos and Ji, 2011], and dimensionality reduction [Sclove, 2021, Tomarchio and Punzo, 2025], to biology [Susko and Roger, 2020], neuroscience [Penny, 2012], and dynamical systems [Thanasutives et al., 2024]. In particular, for regular models, for which the smoothness in the parameter space implies identifiability and the existence of the Fisher information metric [Liu and Suzuki, 2025], the celebrated Bayesian information criterion (BIC) asymptotically approximates the log marginal likelihood and thus carries statistical meaning for posterior inference. Through the Laplace approximation [Schwarz, 1978], the BIC has an explicit formula we can compute directly from the input data.

Singular models: Regularity assumptions are generically violated in deep learning due to the overparameterization of modern architectures [Wei et al., 2022, Liu and Suzuki, 2025]. Hence, for non-regular models, the various statistical quantities are estimated using approximate and costly methods such as sampling [Liu and Suzuki, 2025, Newman, 2026, Lau et al., 2025]. The real log canonical threshold (RLCT) is a key quantity from algebraic geometry that characterizes the asymptotic behavior of the log marginal likelihood for non-regular models, generalizing the role

of model dimension in the BIC [Watanabe, 2009]. Unlike the explicit BIC formula, computing the RLCT requires resolving singularities in the parameter space; a very challenging computational and mathematical problem.

Polynomial neural networks and identifiability: A class of neural networks whose activation functions are monomials $\sigma(x) = x^r$, coined *polynomial neural networks* (PNNs), are gaining interest for their theoretical properties. Their appeal stems from the fact that a PNN defines an *algebraic* map from weights to functions: the composition of monomial activations and linear layers is a polynomial, so the full machinery of algebraic geometry applies. This structure makes it possible to characterize expressivity via algebraic dimension [Kileel et al., 2019] and degree [Kubjas et al., 2024], and gives rise to a rich theory of identifiability [Usevich et al., 2025, Shahverdi et al., 2025]. In particular, the asymptotic behavior of the log marginal likelihood, and hence the identifiability of the model under Bayesian inference, is entirely governed by the RLCT [Watanabe, 2009].

Our Contribution: We propose, to our knowledge, the first deterministic algorithm that exactly computes the RLCT for any polynomial of two variables, and derive an upper bound on its arithmetic complexity (Prop. 2.7, Thm. 2.11, and Thm. 2.12). As an application, we compute exact RLCT for polynomial neural networks (PNNs) with repeated weights of increasing depth. We show that the effective model complexity, as measured by the RLCT, can counter-intuitively decrease with the number of layers, implying that already in the two-dimensional case, this gives rise to a non-trivial theory of identifiability. We advocate that such exact algorithms are new tools for the study of loss landscapes in learning theory.

Notation Let $\mathbb{K}$ be a field of characteristic zero. We let $\bar{\mathbb{K}}$ be the algebraic closure of $\mathbb{K}$. We denote by $\mathbb{K}[x, y]$ the ring of polynomials in two variables. We let $\mathbb{K}[[x]]$ be the ring of formal power series, $\mathbb{K}((x))$ the field of Laurent series and $\mathbb{K}\langle\langle x \rangle\rangle$ the field of Puiseux series with coefficients in $\mathbb{K}$. For an integer n and a subset $S \subset \mathbb{R}^n$, we let $\text{Conv}(S)$ be the convex hull of S . For any two subsets $A \subseteq \mathbb{R}^n$ and $B \subseteq \mathbb{R}^n$, we let $A \oplus B := \{a + b \mid a \in A, b \in B\}$ be their Minkowski sum. For any $n \in \mathbb{N}_{>0}$, we let $[n] = \{1, \dots, n\}$. We use bold roman lowercase letters for vectors in $\mathbb{R}^2$.

1.1 Background

Information Criteria for Model Selection Information criteria are model selection methods in statistics and machine learning that balance predictive accuracy against model complexity. A celebrated example of such a criterion is the Bayesian information criterion (BIC) [Schwarz, 1978], which emanates from the asymptotic behavior of log marginal likelihoods under regularity assumptions on the models. Concretely, consider a family of parametric probability distributions $P(X_1, \dots, X_N \mid \theta, M)$, one for each parameter value $\theta \in \Theta_M$, with possible values in a Borel subset $\Theta_M \subseteq \mathbb{R}^{d_M}$; each family of model M therefore provides the likelihood of a collection of N samples $X_1, \dots, X_N$. For a given prior $\pi_M(\theta)$ over the parameter space Θ_M , the marginal likelihood of the data under model M is $P(x_1, \dots, x_N \mid M) = \int P(x_1, \dots, x_N, \mid \theta, M) \pi_M(\theta) d\theta$. Under appropriate regularity assumptions [Schwarz, 1978, Konishi and Kitagawa, 2008], we can approximate that integral, for large sample size n , as follows:

$$-\log P(x_1, \dots, x_N \mid M) \approx -\log P(x_1, \dots, x_N \mid \hat{\theta}_M, M) + \frac{d_M}{2} \log N, \quad (\text{BIC})$$

where $\hat{\theta}_M$ is the maximum-likelihood estimator (MLE) under model M . The right hand side of Eq. BIC is the expression of the celebrated Bayesian Information Criterion (BIC); it penalizes model complexity (larger d_M) while rewarding goodness of fit. Several other criteria that balance model fit and model complexity are available, such as the Akaike Information Criterion (AIC), the Generalized Information Criterion (GIC), and the Takeuchi Information Criterion (TIC), as well as Bayesian extensions like Akaike’s Bayesian Information Criterion (ABIC).

However, regularity conditions are not satisfied for deep learning architectures [Wei et al., 2022, Watanabe, 2009]. A false asymptotic estimate of the log marginal likelihood, and more generally of the posterior, given a prior on models, can lead to erroneous decisions in model selection [Drton and Plummer, 2017], and so corrections to the BIC have been proposed. The widely applicable Bayesian information criterion (WBIC) relies on the correct asymptotics of the log marginal likelihood,

$$-\log P(x_1, \dots, x_N \mid M) \approx -\log P(x_1, \dots, x_N \mid \hat{\theta}, M) + \lambda \log N, \quad (\text{WBIC})$$

where λ is called the *learning coefficient*. The WBIC is asymptotically consistent in selecting the most parsimonious model [Drton and Plummer, 2017].

From Regular to Singular Models To understand when the BIC is sufficient and when WBIC is needed, we formalize the notion of regularity. The asymptotic expansion of the log marginal likelihood requires it to be re-expressed as follows,

$$P(x_1, \dots, x_N \mid M) = \int e^{N \cdot \frac{1}{N} \sum_i \ln P(x_i \mid \theta, M)} \pi_M(\theta) d\theta.$$

When the sample size N tends to infinity, we get

$$\frac{1}{N} \sum_{i=1}^N \ln P(x_i \mid \theta, M) \xrightarrow{N \rightarrow \infty} \mathbb{E}_{X \sim P} [\ln P(X \mid \theta, M)] \quad P \text{ a.s.},$$

where P is the distribution from which we draw the iid samples $x_1, \dots, x_N$ and $X \sim P$. Up to a term that does not depend on θ , the previous limit is the negative of the Kullback Leibler divergence,

$$D_{KL}(P \parallel P_\theta) = -\mathbb{E}_{X \sim P} [\ln P(X \mid \theta, M)] + \mathbb{E}_{X \sim P} [\ln P(X)]$$

If there exists $\rho \in \Theta$ such that $P(X) = P(X \mid \rho, M)$, then the function $g: \theta \mapsto D_{KL}(P_\rho \parallel P_\theta)$ is minimized at $\theta^* = \rho$ with $g(\theta^*) = 0$; we say that such a model is *realizable*. If the map $\Theta_M \ni \theta \mapsto P(X_1, \dots, X_N \mid \theta, M)$ is injective, then we say that the model is *identifiable*. Moreover, if the Hessian of g at θ is positive definite at every $\theta \in \Theta_M$, we say that the model itself is *positive definite*. To make it simple, if a model is injective and positive definite, it is said to be *regular*. In the regular case, BIC provides the correct asymptotic expansion of the log marginal likelihood under standard regularity conditions. If a model is not regular, then it is called *singular*. In the singular case, WBIC is the correct expression. In this setting the learning coefficient is the RLCT, as we will explain just after. We now introduce the RLCT and review state-of-the-art methods and assumptions for its computation, which stem from the study of singularities of analytic functions.

Real Log Canonical Threshold If the log-likelihood is a real analytic function, the learning coefficient coincides with the *real log canonical threshold* (RLCT) of the Kullback–Leibler divergence at the realizable model, i.e., the true model that generated the samples. The RLCT is a purely algebraic quantity that is well-studied in singularity theory.

In statistical learning, the RLCT of the Kullback–Leibler divergence of a model-truth-prior triplet is the learning coefficient, and its local version is the local learning coefficient (LLC). The global learning coefficient determines the asymptotic behavior of the log marginal likelihood, while the local learning coefficient characterizes the learning dynamics near specific parameter configurations [Lau et al., 2025]. We now give a formal definition.

Definition 1.1 (Real Log Canonical Threshold). Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a real analytic function defined on an open set $O \subset \mathbb{R}^n$. Let C be a compact subset of O . Then, for each $x \in C$ such that $f(x) = 0$ there exists, by Theorem A.1, $W \subseteq \mathbb{R}^n$, an open set containing $\mathbf{0}$, an n -dimensional real analytic manifold U , and a real analytic map $\rho: U \rightarrow W$, such that:

$$f(\rho(u)) - x = \pm u_1^{k_1} \cdots u_n^{k_n}, \quad \text{and} \quad \text{Jac}(\rho) = b(u) u_1^{h_1} \cdots u_n^{h_n}.$$

The local real log canonical threshold at x is given by:

$$\text{RLCT}_x(f) = \min_{1 \leq i \leq n} \frac{h_i + 1}{k_i}.$$

The global RLCT is the minimum local RLCT over all points of C , that is to say,

$$\text{RLCT}(f) := \min_{x \in C} \text{RLCT}_x(f).$$

Note that the RLCT is a rational number. The RLCT is defined using a real analytic birational map, called the resolution of singularities, which, roughly speaking, represents f locally as a normal crossing function. The existence of such a map for real analytic functions was famously proved by

Hironaka [1964]. We recall this theorem in Appendix A. Although the literature on RLCTs remains sparse, as opposed to that of its complex counterpart, the log canonical threshold (LCT) [Mustata, 2012], its algebraic properties have been studied in [Lin, 2011, Chapter 4] and Saito [2007], and were used to classify real hyperplane singularities in Kosta and Windisch [2024]. While Hironaka’s theorem guarantees the existence of a resolution of singularities, computing this resolution explicitly is notoriously difficult and generally impractical [Bierstone et al., 2011]. Our work is motivated by this computational bottleneck.

1.2 Prior Work

The computation of learning coefficients (RLCTs of the Kullback-Leibler divergence for a given sample, model, and generating distribution) remains an active area of research. We review three main categories of work related to this problem: model-specific theoretical results, sampling-based estimation, and algebraic methods for computing RLCTs, with a particular focus on the two-dimensional case where models are parameterized by two real numbers $\theta \in \mathbb{R}^2$.

Learning coefficients have been computed exactly for a range of statistical models through dedicated theoretical analysis. These include mixture models [Yamazaki and Watanabe, 2003], three-layered neural perceptrons [Aoyagi et al., 2005], restricted Boltzmann machines [Aoyagi, 2013], as well as Bayesian networks [Rusakov and Geiger, 2005]. More recently, learning coefficients for deep linear networks of arbitrary depth [Aoyagi, 2024] and factor analysis models [Drton et al., 2025] have also been calculated. Lin [2011] used a Newton polygon based method to compute local learning coefficients, when the Newton polygon satisfies strong geometric constraints. Our work removes these restrictions for the two-dimensional case, providing a general algorithm for arbitrary 2D polynomials.

When exact theoretical results are unavailable, the local learning coefficient can be estimated via sampling methods. Lau et al. [2025] introduced a scalable estimator $\hat{\lambda}^{\text{SGLD}}(\theta^*)$ for the LLC, based on stochastic gradient Langevin dynamics (SGLD) [Welling and Teh, 2011] sampling of the posterior distribution and on Watanabe’s widely applicable Bayesian Information Criterion (WBIC) [Watanabe, 2013]. While SGLD-based estimation has shown empirical success, it provides no theoretical guarantees for sampler convergence [Hitchcock and Hoogland, 2025]. In the absence of known values of the learning coefficient, measuring the convergence of SGLD chains can be difficult and calibrating their hyperparameters can be costly [Vehtari et al., 2021].

For analytic functions in two variables, a line of work beginning with Varchenko [1976] and continuing through Phong et al. [1999] and Collins [2018] established that the local RLCT of a polynomial $f(x, y)$ can be computed using the geometry of the Newton polygons of certain transformations $\tilde{f}(x, y)$ of $f(x, y)$ (see Appendix C). [Paemurru, 2024, Remark 3.11] proposes an algorithmic method, based on Boehm et al. [2020], but it does not terminate for certain polynomial classes (see Section) and lacks complexity bounds where it does. We resolve these issues by adapting the proofs of Phong et al. [1999], Collins [2018] to show that finitely many transformations of $f(x, y)$ suffice to compute the local RLCT of any bivariate polynomial. Our Algorithm 1 computes the exact local RLCT and terminates for any $f(x, y)$, with an explicit upper bound on the number of steps.

2 Theoretical results

In this section, we present our main theoretical results. We present an algorithm (Algorithm 1) to compute the local RLCT of a bivariate polynomial at the origin and we also give a quadratic bound (Theorem 2.12) on its arithmetic complexity, depending on its degree. We sketch the proofs of the various results that support the correctness and the complexity bound of the algorithm. The full proofs appear in the Appendix F. The algorithm combines geometric and algebraic techniques and is based on properties of the Newton polygon of a bivariate polynomial (Appendix C).

2.1 Preliminaries: Newton Polygons and RLCTs

Computing the resolution of singularities of a given analytic function or polynomial is notoriously expensive [Bierstone et al., 2011]. Thus it is desirable to find alternative ways to compute the RLCT. In Varchenko [1976], Varchenko introduced a method to compute the local RLCT of an analytic function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ at an isolated singularity, based on the geometry of its Newton polygon $\mathcal{N}(f)$. This method was later generalized to arbitrary singularities [Phong et al., 1999, Theorem 5].

A modern treatment is given in Collins [2018]. The core idea of the above results is to construct an RLCT-preserving automorphism $\Phi : \mathbb{K}\{x, y\} \rightarrow \mathbb{K}\{x, y\}$, such that the RLCT of an analytic function $f(x, y)$ can be read on the Newton polygon $\mathcal{N}(\Phi(f))$. The main theoretical contribution of our work is to make Varchenko's method effective. Indeed, in some cases, Varchenko's method may not terminate.

We circumvent this issue by identifying precisely the cases where Varchenko's method fails to terminate, and handle them separately. We first define the Newton polygon of a function f and its corresponding Newton distance δ_f .

Definition 2.1 (Newton Polygon [Casas-Alvero, 2000]). Let $\mathbb{K}$ be a field of characteristic zero. Let $f \in \mathbb{K}[[x, y]]$ be a formal power series in two variables:

$$f(x, y) = \sum_{\alpha, \beta=0}^{\infty} c_{\alpha\beta} x^{\alpha} y^{\beta} \in \mathbb{K}[[x, y]].$$

Let $D(f) = \{(\alpha, \beta) \mid c_{\alpha\beta} \neq 0\} \subseteq \mathbb{N}^2$ be the support of f . The Newton polygon, $\mathcal{N}(f)$, is obtained by attaching a copy of the positive quadrant $\mathbb{R}_{\geq 0}^2$ to each point of $D(f)$ and considering the convex hull of the union; that is $\mathcal{N}(f) := \text{Conv}(D(f) \oplus \mathbb{R}_{\geq 0}^2)$, where $\oplus$ denotes the Minkowski sum.

The Newton distance is a positive rational number associated to the Newton polygon $\mathcal{N}(f)$ of $f(x, y)$, which measures how far the Newton polygon is from the origin.

Definition 2.2 (Main face and Newton distance). The face Δ at which the diagonal $\mathcal{D} = \{(\alpha, \beta) \in \mathbb{R}_{\geq 0}^2 \mid \alpha = \beta\}$ intersects the boundary $\partial\mathcal{N}(f)$ of the Newton polygon is called the *main face* of $f(x, y)$. Let $\mathbf{p} = (\delta, \delta) \in \mathbb{Q}_{\geq 0}^2$ be the point of intersection of $\mathcal{D}$ with $\partial\mathcal{N}(f)$. Then, $\delta_f := \delta \in \mathbb{Q}_{\geq 0}$ is the *Newton distance* of f .

We remind the reader of the notion of right equivalence of power series.

Definition 2.3 (Right equivalence of power series). Let $f, g \in \mathbb{K}[[x, y]]$ be formal power series. We say that f and g are right-equivalent if there exists an isomorphism $\Phi : \mathbb{K}[[x, y]] \rightarrow \mathbb{K}[[x, y]]$, such that $f \circ \Phi = g$.

Next, we state the normalization condition on the Newton polygon $\mathcal{N}(f)$ of $f(x, y)$. If a polynomial or analytic function $f(x, y)$ satisfies the normalization condition, then $\text{RLCT}_0(f) = \frac{1}{\delta_f}$.

Definition 2.4 (Normalization condition Phong et al. [1999], Collins [2018]). Let $f(x, y) \in \mathbb{R}\{x, y\}$, and let $\mathcal{N}(f)$ be its Newton polygon. Let Δ_i for $i \in [K]$ be the facets of $\mathcal{N}(f)$. Let $F_{\Delta_i}(z)$ be its facet polynomials and let a_{ij} and m_{ij} for $j \in [r_i]$ be its distinct roots, and their corresponding multiplicities, respectively. We say that Δ_i is normalized if $\delta_f \geq m_{ij}$ for all $j \in [r_i]$. We say that $f(x, y)$ is normalized if the facet Δ_i of its Newton polygon $\mathcal{N}(f)$ is normalized for all $i \in [K]$.

Proposition 2.5 (Collins [2018]). Let $f \in \mathbb{K}\{x, y\}$. If $f(x, y)$ is right equivalent to a normalized power series $g(x, y)$, then $\text{RLCT}_0(f) = \frac{1}{\delta_g}$.

The following proposition plays an important role in our work. The version that we present appears in [Phong et al., 1999, Theorem 5] and [Collins, 2018, Theorem 1.4]; a first version for real isolated plane curve singularities appears in [Varchenko, 1976, Theorem 0.6]. It guarantees that for any bivariate analytic function f , one can construct a right equivalence Φ , such that f is normalized. For completeness, we present a self-contained proof in the Appendix E.1, where we also show that $P_f(x)$ or $Q_f(y)$ are truncations of Puiseux roots of $f(x, y)$.

Proposition 2.6 (Phong et al. [1999], Collins [2018]). Any analytic function $f \in \mathbb{K}\{x, y\}$ is right equivalent to a normalized power series. Moreover, this right equivalence is given by a change of variable $\Phi : \mathbb{K}[[x, y]] \rightarrow \mathbb{K}[[x, y]]$, $(x, y) \mapsto (x, y - P_f(x))$ or $\Phi : \mathbb{K}[[x, y]] \rightarrow \mathbb{K}[[x, y]]$, $(x, y) \mapsto (x - Q_f(y), y)$, where $P_f(x)$ and $Q_f(y)$ are power series of strictly positive order in $\mathbb{K}[[x]]$ and $\mathbb{K}[[y]]$ respectively.

Proposition 2.6 suggests that the local RLCT of $f(x, y)$ is computable up to arbitrary precision, but it is not effective, as $P_f(x)$ can have infinitely many terms. Our work resolves this issue.

2.2 Main Results

In this work, we show that it is sufficient to compute $P_f(x)$ up to a *finite* degree in order to compute the local RLCT of a polynomial $f(x, y)$ at 0. We obtain a computable bound on this degree, and thus

on the maximum number of iterations necessary for computing the local RLCT. We then provide an exact, effective algorithm, which takes as input any polynomial $f(x, y) \in \mathbb{Q}[x, y]$ and outputs $\text{RLCT}_0(f) \in \mathbb{Q}$. Without loss of generality, we henceforth focus only on the situation where $P_f(x)$ gives the required right equivalence. Otherwise, one can consider the change of variable $x \mapsto y$, $y \mapsto x$. We show that if $P_f(x) \in \mathbb{K}[[x]]$ is such that $f(x, y - P_f(x))$ is normalized, then $P_f(x)$ is unique. Furthermore, if the coefficients of $f(x, y)$ are rational, then so are the coefficients of $P_f(x)$. The proof of the following lemma in Appendix F.1.

Proposition 2.7. *For any unnormalized analytic function $f(x, y) \in \mathbb{K}\{x, y\}$, the power series $P_f(x)$ computed in the proof of Prop. 2.6 is unique. Moreover, if $\mathbb{K} = \mathbb{Q}$, then $P_f(x) \in \mathbb{Q}[[x]]$.*

We can thus speak of the normalizing power series of a bivariate polynomial.

Definition 2.8 (Normalizing power series). Let $f(x, y)$ be an analytic function. We call a power series $P_f(x)$ the *normalizing power series* of $f(x, y)$ if $\tilde{f}(x, y) = f(x, y - P_f(x))$ is normalized.

Remark 2.9. Recall that a y -root of $f(x, y)$ is an element $\phi(x)$ of the algebraic closure of $\mathbb{K}[x]$, such that $f(x, \phi(x)) = 0$ (see Appendix D). In the course of the proof of Proposition 2.6, it is shown that $P_f(x)$ corresponds either:

- (1) to a finite truncation of some Puiseux root of strictly positive order of $f(x, y)$, or,
- (2) to a power series of strictly positive order $\phi(x) \in \bar{\mathbb{K}}[[x]] \setminus \mathbb{K}[x]$ which is a y -root of $f(x, y)$.

Using the Newton-Puiseux algorithm, $P_f(x)$ can thus be computed entirely, in the first case, or up to an arbitrary number of terms in the second case.

We first define the finite part $P_f^{\text{fin}}(x)$ of $P_f(x)$, where we use the notion of singular parts of Puiseux series (Definition D.3). For the remainder of the section, we let $\mathbb{K} = \mathbb{Q}$.

Definition 2.10 (Finite part of a normalizing power series). Let $f(x, y) \in \mathbb{Q}[x, y]$, with normalizing power series $P_f(x)$. If there exists a y -root $\phi(x)$ of $f(x, y)$, such that $P_f(x) = \phi(x)$, then we define the finite part $P_f^{\text{fin}}(x)$ of $P_f(x)$ to be the singular part (Definition D.3) of $P_f(x)$; that is $P_f^{\text{fin}}(x) := S_{P_f}(x)$. Otherwise, we let $P_f^{\text{fin}}(x) := P_f(x)$.

The following result justifies the algorithm we propose in the next section. It establishes that a finite part of $P_f(x)$, which distinguishes it from all the other Puiseux roots $\phi(x)$ of strictly positive order, is sufficient to compute the local RLCT of a polynomial $f(x, y)$.

Theorem 2.11. *Let $f(x, y) \in \mathbb{Q}[x, y]$, with normalizing power series $P_f(x) \in \mathbb{Q}[[x]]$. Let $P_f^{\text{fin}}(x)$ be its finite part. Let $\tilde{f}(x, y) = f(x, y - P_f^{\text{fin}}(x))$. Let Δ be the main face of $\mathcal{N}(f)$. If $\tilde{f}$ is normalized, then $\text{RLCT}_0(f) = \frac{1}{\delta_{\tilde{f}}}$. If not, $\text{RLCT}_0(f) = \frac{1}{m}$, where $m > \delta_f$ is the largest multiplicity of the roots of $\tilde{F}_{\Delta}(z)$.*

Proof. This theorem follows from the unicity of $P_f(x)$, and from the maximality of the degree of singular parts of Puiseux roots of polynomials. We defer the detailed proof to Appendix F.2. $\square$

This result tells us precisely under which conditions Varchenko's method fails to terminate, and recovers the RLCT even in those pathological cases. Our second theoretical result generalizes a univariate root separation bound [Tsigaridas and Emiris, 2008], to an upper bound on the degree of $P_f^{\text{fin}}(x)$. This allows us to define a stopping criterion for our algorithm and thus is also a bound on the number of steps that our algorithm takes before terminating.

Theorem 2.12. *Let $f(x, y) \in \mathbb{Q}[x, y]$, such that $\deg_y f = d_y$ and $\deg_x f = d_x$. Consider the finite part of its normalizing power series, $P_f^{\text{fin}}(x)$. Let d be the degree of $P_f^{\text{fin}}(x)$. Then $d < (d_y + \frac{1}{2})d_x$.*

Sketch of Proof. The proof in full detail can be found in Appendix F.3. Suppose that the finite part $P_f^{\text{fin}}(x)$ is shared by at least two Puiseux roots of $f(x, y)$, say $\phi(x)$ and $\psi(x)$. The order of the difference between ϕ and ψ , $\text{ord}(\phi(x) - \psi(x))$ is an upper bound on the degree of $P_f^{\text{fin}}(x)$. We are thus interested in finding upper bounds on the order of the differences of the roots of $f(x, y)$, based on the degree in x and y of $f(x, y)$. This is analogous to the root separation problem in the univariate

case: finding a lower bound on the absolute difference between the roots of a given polynomial $f(z)$. We generalize the approach of Tsigaridas and Emiris [2008], which lower bounds the root separation of a polynomial $g(z) \in \mathbb{Z}[z]$, using the discriminant of g . We argue that the degree in x of the discriminant of $f(x, y)$ gives a similar upper bound in the bivariate case. $\square$

Theorem 2.12 provides a bound on the number of steps necessary to implement the check in Theorem 2.11. Combining Theorem 2.11 and Theorem 2.12, we are now ready to construct our algorithm.

3 An exact algorithm to compute the RLCT of 2D polynomials

Algorithm 1: ComputeRLCT2D

Input: A polynomial $f(x, y) \in \mathbb{Q}[x, y]$.

Output: $\text{RLCT}_0(f)$

```

1  $d_y \leftarrow \deg_y f$ 
2  $d_x \leftarrow \deg_x f$ 
3  $B \leftarrow (d_y + \frac{1}{2})d_x$ 
4  $d \leftarrow 0$ 
5 while  $B > 0$  do
6   Compute the Newton polygon  $\mathcal{N}(f)$  of  $f$ 
7   Compute the Newton distance  $\delta_f$  of  $f$ 
8   if  $\text{Normalized}(f, \mathcal{N}(f), \delta_f) = \text{True}$  then
9     return  $\frac{1}{\delta_f}$ 
10  else if  $\text{Normalized}(f, \mathcal{N}(f), \delta_f) = (b, p, m)$  then
11     $f(x, y) \leftarrow f(x, y - bx^p)$ 
12     $B \leftarrow B - (p - d)$ 
13     $d \leftarrow p$ 
14  end
15 end
16 if  $\text{Normalized}(f, \mathcal{N}(f), \delta_f) = \text{True}$  then
17   return  $\frac{1}{\delta_f}$ 
18 else if  $\text{Normalized}(f, \mathcal{N}(f), \delta_f) = (b, p, m)$  then
19   return  $\frac{1}{m}$ 
20 end

```

In this section, we propose an algorithm for computing the local RLCT at the origin of any polynomial $f \in \mathbb{Q}[x, y]$ (Algorithm 1). This algorithm implements the constructive proof of the existence of a normalizing power series, as described in Phong et al. [1999], Collins [2018], but includes a stopping criterion derived from Theorem 2.12, such that the normalizing power series is only computed up to $P_f^{\text{fin}}(x)$. By Theorem 2.11, this is sufficient to compute $\text{RLCT}_0(f)$. The definition is self-contained, except for the `Normalized` subroutine, which checks if a polynomial f satisfies the normalization condition (Definition 2.4). The definition of the `Normalized` subroutine is deferred to Appendix H. We prove the correctness of Algorithm 1. The proof is deferred to Appendix G.1.

Theorem 3.1 (Correctness). *Let $f(x, y) \in \mathbb{Q}[x, y]$. Then Algorithm 1 computes the real log canonical threshold at the origin of $f(x, y)$.*

We obtain upper bounds on the number of iterations required to compute the RLCT using Algorithm 1.

Corollary 3.2. *Let $f(x, y) \in \mathbb{Q}[x, y]$, such that $\deg_y f = d_y$ and $\deg_x f = d_x$. Then, Algorithm 1 computes the $\text{RLCT}_0(f)$ in less than $(d_y + \frac{1}{2})d_x$ iterations.*

Proof. Theorem 3.1 asserts that Algorithm 1 computes the $\text{RLCT}_0(f)$. The number of iterations of the while loop (lines 14–15) is upper bounded by $B = (d_y + \frac{1}{2})d_x$. $\square$

4 Application to polynomial models

We demonstrate the applicability of our algorithm by computing local learning coefficients of biparametric models. We define an equivalence relation on the space of real functions which allows us to apply our algorithm, even when the Kullback-Leibler distance of the considered model is not polynomial. We illustrate this for Polynomial Neural Networks in Section 4.1.

4.1 Polynomial Neural Networks

Polynomial neural networks (PNNs) are a class of neural network models whose activations are monomial functions, which recently have shown state-of-the-art performance on a variety of tasks, from image generation and classification [Chrysos et al., 2020, Huang et al., 2003], to trading signals forecasting [Ghazali et al., 2011], signal representation [Yang et al., 2022] and to the resolution of inverse problems in physics [Bu and Karpatne, 2021], among others.

Definition 4.1 (Polynomial Neural Network [Kubjas et al., 2024]). A (bias-less) *polynomial neural network* f_θ , with architecture $d = (d_0, \dots, d_L)$ is a function $f_\theta : \mathbb{R}^{d_0} \rightarrow \mathbb{R}^{d_L}$, defined as:

$$f_\theta = W_L \circ \sigma_{L-1} \circ W_{L-1} \circ \sigma_{L-2} \circ \dots \circ \sigma_1 \circ W_1,$$

where $W_i \in \mathbb{R}^{d_i \times d_{i-1}}$ are linear maps, and the monomial activation functions $\sigma_i : \mathbb{R}^n \rightarrow \mathbb{R}^n$ act component-wise and are given by:

$$\sigma_i(x) = (x_1^r, \dots, x_n^r).$$

The integer r is called the activation degree of the network. The parameters θ are given by the entries of the matrices (or weights) $W_1, \dots, W_L$.

As PNNs define an algebraic map from weights to polynomials, their polynomial nature makes them particularly amenable to the tools of algebraic geometry. For example, authors have used algebraic dimensions [Kileel et al., 2019], algebraic degrees [Kubjas et al., 2024], singularity theory [Shahverdi et al., 2025], and low-rank tensor decompositions [Usevich et al., 2025], to study the expressivity of PNNs, to bound their number of learnable functions, characterize subnetworks and analyze their identifiability, respectively. In this section, we apply Algorithm 1 to a regression model induced from a PNN with repeated weights, and compute its local learning coefficient at the origin.

4.2 Experimental Setup

We consider PNNs of varying depth L , with parameters $\theta = (\theta_1, \theta_2) \in \Theta \subset \mathbb{R}^2$ and repeat L times the weight

$$W = \begin{pmatrix} p_{11}(\theta_1, \theta_2) & p_{12}(\theta_1, \theta_2) \\ p_{12}(\theta_1, \theta_2) & p_{22}(\theta_1, \theta_2) \end{pmatrix} \in \mathbb{Q}[\theta_1, \theta_2]^{2 \times 2}.$$

We fix the activation degree of the network to be $r = 2$, such that:

$$f_\theta(x_1, x_2) = W \circ \sigma_{L-1} \circ W \circ \sigma_{L-2} \circ \dots \circ \sigma_1 \circ W(x_1, x_2). \quad (1)$$

Write $x = (x_1, x_2)$. We will furthermore require that, for any polynomial entry of the weight $p_{ij}(\theta_1, \theta_2)$, that $p_{ij}(0, 0) = 0$. Letting $\theta^* = (0, 0)$, we observe that $f_{\theta^*}(x) = (0, 0)$.

As in [Wei et al., 2022, Aoyagi, 2024], we focus on the regression task. We consider the family of models:

$$\forall y, x \in \mathbb{R}^2 \quad p(y | x, \theta) = \frac{1}{2\pi} \exp\left(-\frac{1}{2} \|y - f_\theta(x)\|^2\right).$$

Let the parameter space Θ be a compact subset of $\mathbb{R}^2$, containing the origin. Let $q(y | x)$ be the true generating distribution of the data. Assume that a continuous distribution $q(x)$ on $\mathbb{R}^2$ is fixed, with respect to which $p(x, y) = p(y|x)q(x)$ and $q(x, y) = q(y|x)q(x)$. We assume that $p(y | x, \theta)$ is a realizable model, that is to say, the set $\Theta_0 = \{\theta \in \Theta \mid p(y | x, \theta) = q(y | x)\}$ is non-empty. Furthermore, let us assume that $(0, 0) = \theta^* \in \Theta_0$. We let $K(\theta) = \mathbb{E}_{q(x)} [D_{KL}(p(y | x, \theta) \parallel q(y | x))]$. Then, by straightforward calculations [Wei et al., 2022, Appendix A.1],

$$K(\theta) = \int_X \|f_\theta(x) - f_{\theta^*}(x)\|^2 q(x) dx = \int_X \|f_\theta(x)\|^2 q(x) dx.$$

Recall that the local learning coefficient at θ_0 of $p(y \mid x, \theta_0)$ is none other than the local real log canonical threshold of $K(\theta)$ at θ_0 . However, $K(\theta)$ is not a polynomial, and we cannot apply Algorithm 1. This is resolved by considering an auxiliary "contact-equivalent" polynomial $H(\theta)$ whose local RLCT at the origin is equal to that of $K(\theta)$ (Proposition B.3). Appendix B discusses contact equivalence and how to compute contact equivalent polynomials. Proposition B.5 is readily applicable to $K(\theta)$.

4.3 Examples

Let $W = \begin{pmatrix} \theta_1 + \theta_2 & \theta_1^2 \\ \theta_1^2 & \theta_2^2 \end{pmatrix}$. For each $1 \leq L \leq 5$ and $2 \leq r \leq 4$, we compute $H_{L,r}(\theta)$, the sum-of-squares polynomial contact equivalent to $K_{L,r}(\theta)$, the Kullback–Leibler distance of the depth- L , activation- r PNN f_θ of Equation 1. The $H_{L,r}(\theta)$ are not normalized, so Algorithm 1 takes at least one step before outputting $\lambda_{L,r} = \text{RLCT}_0(H_{L,r})$; such degenerate polynomials lie outside the reach of prior Newton polygon methods like [Lin, 2011, Section 4.2.1].

Table 1: Local RLCT at the origin of $H_{L,r}(\theta)$: exact rationals $\lambda_{L,r}$ from Algorithm 1 (SageMath, Intel Core Ultra 7) versus SGLD estimates $\hat{\lambda}^{\text{SGLD}}$ (absolute error in parentheses; $\gamma = 1$, $\epsilon = 10^{-5}$, $C = 5$ chains, $T = 10,000$; Google Colab, 2 vCPUs). Exp is the time to expand $H_{L,r}(\theta)$; Alg. 1 is Algorithm 1 alone. NaN marks numerical instability; >1 hr marks runs that did not finish for Algorithm 1.

		$L = 1$	$L = 2$	$L = 3$	$L = 4$	$L = 5$
$r = 2$	$\lambda_{L,r}$	3/4	1/4	3/28	1/20	3/124
	Exp	0.001s	0.004s	0.031s	2.09s	153.7s
	Alg. 1	0.045s	0.059s	0.271s	3.17s	66.4s
	Total	0.047s	0.063s	0.302s	5.26s	220.2s
	$\hat{\lambda}^{\text{SGLD}}$	.677 (.072)	.154 (.095)	.038 (.068)	.014 (.035)	.006 (.017)
	Time (SGLD)	461s	430s	421s	473s	581s
$r = 3$	$\lambda_{L,r}$	3/4	3/16	3/52	3/160	>1 hr
	Exp	0.001s	0.007s	1.92s	868s	–
	Alg. 1	0.045s	0.141s	2.73s	304s	–
	Total	0.047s	0.148s	4.65s	1173s	–
	$\hat{\lambda}^{\text{SGLD}}$	.677 (.072)	.081 (.106)	.025 (.033)	.007 (.011)	NaN
	Time (SGLD)	576s	474s	485s	495s	–
$r = 4$	$\lambda_{L,r}$	3/4	3/20	1/28	>1 hr	>1 hr
	Exp	0.001s	0.004s	6.13s	–	–
	Alg. 1	0.045s	0.079s	12.5s	–	–
	Total	0.047s	0.083s	18.6s	–	–
	$\hat{\lambda}^{\text{SGLD}}$	.677 (.072)	.093 (.142)	.019 (.017)	NaN	NaN
	Time (SGLD)	456s	431s	443s	–	–

The pre-processing step of expanding $H_{L,r}(\theta)$ becomes the computational bottleneck at greater depth and activation degree, but is independent of the RLCT computation itself, which remains competitive against SGLD sampling and requires no hyperparameter tuning. The local learning coefficients decrease as a function of the depth L and the activation degree r , suggesting that PNNs with repeated weights get more degenerate when the number of layers is increased: this is likely because repeated weights increase the multiplicity of the origin as a zero of $K_{L,r}(\theta)$. As the number of layers increase, so does the degree and the size of the coefficients $H_{L,r}(\theta)$, leading to increased wall-clock time. To run our SGLD experiments, we used Timaeus’ DevInterp library Snell et al. [2026].

5 Discussion

Towards exact algorithms for the learning theory of low-dimensional models: We developed an effective algorithm that computes exactly the real log canonical threshold of any input polynomial $f(x, y) \in \mathbb{Q}[x, y]$ in a number of steps quadratic in $\deg f$, although the tightness of this bound remains open. The algorithm is model-agnostic, broadening the class of two-dimensional models for which local learning coefficients can be computed.

We demonstrated it by computing local learning coefficients of PNNs with repeated weights of increasing depth. Exact values are crucial for model selection in smaller models and can be used to calibrate sampling-based estimators [Lau et al., 2025] for larger models. Effective model complexity, as measured by the RLCT, can decrease with depth - revealing non-trivial identifiability structure already in the two-dimensional case.

Limitations and beyond: Algorithm 1 is limited to biparametric models. As observed in [Phong et al., 1999, Section III], in dimensions > 2 we still do not know how to characterize good coordinate systems with respect to Newton diagrams. A promising approach is that of [Collins et al., 2013], where the normalizing changes of variables are given by multivariate fractional power series. A second limitation is locality: the assumption in this work is that a singular point of $K(\theta)$ is given. For practical applications, such as *internal model selection* (see Chen et al. [2023]), it would require a stratification of the singular locus of f , whose effective computation is studied in e.g. [Helmer and Nanda, 2023].

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A Hironaka’s Resolution of Singularities

Here we recall the statement of Hironaka’s resolution of singularities, following Watanabe [2007].

Theorem A.1 (Resolution of singularities, [Hironaka, 1964]). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a non-constant real analytic function, such that $f(\mathbf{0}) = 0$. Let $V(f) = \{x \in \mathbb{R}^n \mid f(x) = 0\}$. Then there exists $W \subseteq \mathbb{R}^n$ an open set containing $\mathbf{0}$, U an n -dimensional real analytic manifold and $\rho : U \rightarrow W$ a real analytic map, such that the triplet (W, U, ρ) satisfies the following conditions:*

1. ρ is a proper map.
2. Writing $W_0 = V(f) \cap W$ and $U_0 = V(f \circ \rho) \cap U$, $\rho : U \setminus U_0 \rightarrow W \setminus W_0$ is a real analytic isomorphism.
3. For any $p \in U_0$, there exists a local coordinate $(u_1, \dots, u_n)$ of U in which p is the origin and

$$f(\rho(u)) = S u_1^{k_1} \cdots u_n^{k_n}$$

where $S = 1$ or $S = -1$ is a constant, $k_1, \dots, k_n$ are non-negative integers, and the Jacobian of ρ satisfies

$$\text{Jac}(\rho) = b(u) u_1^{h_1} \cdots u_n^{h_n},$$

where $b(u) \neq 0$ is a real analytic function not vanishing at p and $h_1, \dots, h_n$ are non-negative integers.

B Polynomial Contact Equivalence

In this section, a more analytic definition of the local RLCT will be particularly useful.

Definition B.1 (Analytic definition of the real log canonical threshold [Collins, 2018]). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a real analytic function. The local real log canonical threshold at x is defined as:

$$\text{RLCT}_x(f) = \sup \left\{ s \in \mathbb{R}_{\geq 0} \mid \exists \epsilon > 0 \text{ s.t. } \int_{B_\epsilon(x)} |f(x)|^{-s} dx < \infty \right\}$$

Recall that (Section 1.1) the local learning coefficient of a parametric model $p(x \mid \theta)$ at a parameter θ^* is the local RLCT of the Kullback-Leibler distance $K(\theta) = D_{KL}(p(x \mid \theta) \parallel q(x))$ at θ^* , where $q(x)$ is the true data generating distribution. $K(\theta)$ is not necessarily a polynomial function, and we cannot directly apply Algorithm 1. A workaround is to find a polynomial $H(\theta) \in \mathbb{R}[\theta]$, equivalent to $K(\theta)$ in the following sense.

Definition B.2 (Contact equivalence [Ruas, 2020, Definition 5.6.5]). Two real functions $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ are contact equivalent at a point $x_0 \in \mathbb{R}^n$ if there exists a neighbourhood $\mathcal{U}_{x_0}$ of x_0 in $\mathbb{R}^n$ such that, there exists positive constants $c_1, c_2 > 0$, such that for all $x \in \mathcal{U}_{x_0}$, we have:

$$c_1 |g(x)| \leq |f(x)| \leq c_2 |g(x)|$$

and $f(x)g(x) \geq 0$. We write $f(x) \sim g(x)$.

Contact equivalence at a point x_0 preserves the local real log canonical threshold at x_0 .

Proposition B.3. Let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ s.t. $f \sim g$ at x_0 . Then, $\text{RLCT}_{x_0}(f) = \text{RLCT}_{x_0}(g)$.

Proof. For simplicity, let $x_0 = 0$ be the origin in $\mathbb{R}^n$ and let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ be positive functions such that $f \sim g$. Let $\text{RLCT}_0(f) = \lambda$. Since $f \sim g$, there exists a positive constant c such that $f(x) \leq cg(x)$ for all x in some neighborhood of the origin. For all $s \in (0, \lambda)$, there exists $\epsilon > 0$ sufficiently small such that $\infty > c^s \int_{B_\epsilon(0)} f(x)^{-s} dx \geq \int_{B_\epsilon(0)} g(x)^{-s} dx$ and $g(x)^{-s}$ is integrable. Therefore,

$$\left\{ s \in \mathbb{R}_{\geq 0} \mid \exists \epsilon > 0 \text{ s.t. } \int_{B_\epsilon(0)} f(x)^{-s} dx < \infty \right\} \subseteq \left\{ s \in \mathbb{R}_{\geq 0} \mid \exists \epsilon > 0 \text{ s.t. } \int_{B_\epsilon(0)} g(x)^{-s} dx < \infty \right\}$$

and hence $\text{RLCT}_0(f) \leq \text{RLCT}_0(g)$. Now, using the left side inequality of the equivalence relation, we find in the same fashion that $\text{RLCT}_0(f) \geq \text{RLCT}_0(g)$, therefore they must be equal. $\square$

Remark B.4. The real log canonical threshold is an invariant of a broader type of equivalence of functions, namely *bi-Lipschitz $\mathcal{K}$ equivalence* [Bivià-Ausina and Fukui, 2016, Theorem 7.3]. Two function germs (i.e. defined locally around the origin) $f, g : (\mathbb{R}^n, 0) \rightarrow (\mathbb{R}, 0)$ are said to be bi-Lipschitz $\mathcal{K}$ equivalent if there exists a bi-Lipschitz homeomorphism $\varphi : (\mathbb{R}^n, 0) \rightarrow (\mathbb{R}^n, 0)$ and a bi-Lipschitz homeomorphism $\Phi : (\mathbb{R}^n \times \mathbb{R}, 0) \rightarrow (\mathbb{R}^n \times \mathbb{R}, 0)$, $(x, y) \mapsto (\varphi(x), \phi(x, y))$, such that $\Phi(\mathbb{R}^n \times \{0\}) = \mathbb{R}^n \times \{0\}$ and $\phi(x, f(x)) = g(\varphi(x))$ for all x in a neighbourhood of the origin. A weaker equivalence is that of bi-Lipschitz $\mathcal{C}$ equivalence, which requires that $\varphi = \text{id}$. In [Ruas, 2020, Theorem 5.6.7], it is shown that two Lipschitz functions are of the same contact if and only if they are bi-Lipschitz $\mathcal{C}$ equivalent.

The following proposition will be used to construct a polynomial equivalent to the Kullback-Leibler distance of the model, when the latter can be written in a specific form.

Proposition B.5 (Aoyagi [2024], Theorem 1). Let $x \in \mathbb{R}^n$ and let $w \in \mathbb{R}^m$. Using multi-index notation, write $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$. Let

$$h(x, w) = \sum_{0 \leq |\alpha| \leq H} \tilde{h}_\alpha(w) x^\alpha$$

be a polynomial in x , where the coefficients $\tilde{h}_\alpha(w)$ are continuous functions. Let $q(x)$ be a positive, continuous function on $X \subset \mathbb{R}^n$, such that $\int_X q(x) dx > 0$. Define

$$K(w) = \int_X h^2(x, w) q(x) dx.$$

Then $K(w)$ is contact equivalent to the sum of the squared coefficients of $h(x, w)$:

$$K(w) \sim H(w) = \sum_{0 \leq |\alpha| \leq H} \tilde{h}_\alpha^2(w).$$

C Geometry of the Newton polygon

In this section, $\mathbb{K}$ is a field of characteristic zero. We give a short presentation of Newton polygons and Puiseux series; we closely follow [Casas-Alvero, 2000, Chapter 1], where we also refer the reader for further details. Consider a polynomial $f(x, y) = \sum_{\alpha, \beta=0}^{\infty} c_{\alpha\beta} x^\alpha y^\beta \in \mathbb{K}[x, y]$, such that $d_y = \deg_y f$ and $d_x = \deg_x f$. By convention, we define the degree of the zero polynomial to be $\deg 0 = -\infty$. A term of $f(x, y)$ is a monomial scaled by a non-zero constant, such as $c_{\alpha\beta} x^\alpha y^\beta$. An exponent of $f(x, y)$ is a tuple (α, β) such that $c_{\alpha\beta} \neq 0$. We associate to f a convex polygon, $\mathcal{N}(f)$, that encodes geometric information about the exponents that occur in f .

Definition C.1. (Newton Polygon) Let $D(f) = \{(\alpha, \beta) \mid c_{\alpha\beta} \neq 0\} \subseteq \mathbb{N}^2$ be the Newton diagram of $f(x, y)$. We obtain the Newton polygon, $\mathcal{N}(f)$, of f by attaching a copy of the positive quadrant $\mathbb{R}_{\geq 0}^2$ to each point of $D(f)$ and considering the convex hull of the union. In particular, we have

$$\mathcal{N}(f) := \text{Conv}(D(f) \oplus \mathbb{R}_{\geq 0}^2).$$

Let $\mathbf{p}_i = (\alpha_i, \beta_i), i \in [K+1]$ be the vertices of $\mathcal{N}(f)$, ordered from left to right. Let Δ_j , for $0 \leq j \leq K+1$, denote the facets of $\mathcal{N}(f)$, ordered from left to right; then, Δ_0 and Δ_{K+1} are the two non-compact facets (in our case, half-lines) of $\mathcal{N}(f)$. Let $\Delta_i = (\mathbf{p}_i, \mathbf{p}_{i+1}), i \in [K]$ be the compact facets (in our case, segments) of $\mathcal{N}(f)$ with vertices $\mathbf{p}_i$ and $\mathbf{p}_{i+1}$. For a compact facet Δ_i , the height and the width are $h_i = \beta_{i-1} - \beta_i$ and $w_i = \alpha_i - \alpha_{i-1}$, respectively. The slope of Δ_i is given by $s_i = -\frac{q_i}{p_i}$, for p_i and q_i two co-prime positive integers. We call $wt_i = (p_i, q_i)$ the weight of a compact facet Δ_i . The length of a compact facet Δ_i is the integer $l_i = \frac{h_i}{q_i} = \frac{w_i}{p_i}$. For the non-compact facets, we set $l_0 = l_{K+1} = \infty$. Finally, let $\bar{\alpha}$ be the largest integer such that $x^{\bar{\alpha}}$ divides $f(x, y)$ and $\bar{\beta}$ be the largest integer such that $y^{\bar{\beta}}$ divides $f(x, y)$. Then, the height and the width of the Newton polygon of f are $h_f = \beta_0 - \bar{\beta}$ and $w_f = \alpha_k - \bar{\alpha}$, respectively. An illustration of the Newton polygon of a given polynomial is given in Figure 1.

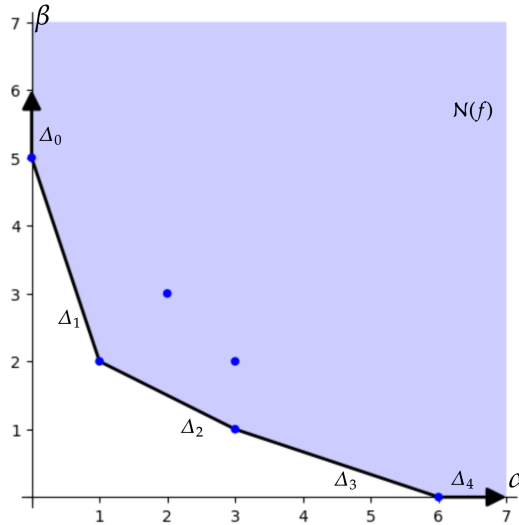


Figure 1: Newton polygon $\mathcal{N}(f)$ of $f(x, y) = y^5 + y^2x + yx^3 + x^6 + y^2x^3 + y^3x^2$, with $K = 3$ compact faces.

For each compact facet Δ_i of weight $wt_i = (p_i, q_i)$, we can consider the restriction of $f(x, y)$ to this facet:

$$f_{\Delta_i}(x, y) := \sum_{(\alpha, \beta) \in D(f) \cap \Delta_i} c_{\alpha\beta} x^\alpha y^\beta \in \mathbb{K}[x, y]. \quad (2)$$

The exponents (α, β) of the restriction f_{Δ_i} , all lie on the lattice points of the segment supporting Δ_i . Hence, we have that $\alpha p_i + \beta q_i = k_i$, for some integer k_i , where $-\frac{q_i}{p_i}$ is the slope of Δ_i . In other words, it is a (p_i, q_i) -weighted homogeneous polynomial. Therefore, f_{Δ_i} corresponds to a univariate polynomial, which we call the *facet polynomial* of Δ_i :

Definition C.2 (Facet polynomial). Let $f(x, y) \in \mathbb{K}[x, y]$, with Newton polygon $\mathcal{N}(f)$. Take a compact facet Δ_i of weight (p_i, q_i) and length l_i . Consider the restriction $f_{\Delta_i}(x, y) = \sum_{(\alpha, \beta) \in D(f) \cap \Delta_i} c_{\alpha\beta} x^\alpha y^\beta$ as above. Then,

$$f_{\Delta_i}(x, y) = x^{\alpha_{i+1}} y^{\beta_{i+1}} \sum_{t=0}^{l_i} c_t (x^{-p_i} y^{q_i})^t,$$

where $c_t = c_{\alpha\beta}$ for $(\alpha, \beta) = (\alpha_{i+1} - p_i t, \beta_{i+1} + q_i t)$. We call the polynomial

$$F_{\Delta_i}(z) := \sum_{t=0}^{l_i} c_t z^t,$$

the *facet polynomial* of Δ_i in $\mathcal{N}(f)$. Note that if Δ is a non-compact facet then $f_{\Delta}(x, y) = x^{\bar{\alpha}} a(y)$ or $f_{\Delta}(x, y) = y^{\bar{\beta}} b(x)$, when Δ is a vertical or horizontal facet, respectively. In this case, we let $F_{\Delta}(z) = z^{\bar{\alpha}}$ or $F_{\Delta}(z) = z^{\bar{\beta}}$, respectively.

Note that, for any facet Δ_i of $\mathcal{N}(f)$, $F_{\Delta_i}(z) \in \mathbb{K}[z]$. Moreover, if Δ_i is compact, then $\deg(F_{\Delta_i}(z)) = l_i$, the length of Δ_i . Let r_i be the number of distinct roots of F_{Δ_i} . We denote by a_{ij} , for $j \in [r_i]$, the distinct roots of F_{Δ_i} , and by m_{ij} , for $j \in [r_i]$, their respective multiplicities.

Definition C.3 (Main face). Let $\mathcal{D} = \{(\alpha, \beta) \in \mathbb{R}_{\geq 0}^2 \mid \alpha = \beta\}$ be the diagonal of the quadrant $\mathbb{R}_{\geq 0}^2$. The *main face* of f is the face of the boundary $\partial \mathcal{N}(f)$ of the Newton polygon, intersected by $\mathcal{D}$.

Definition C.4 (Newton distance). Let $\mathbf{p} = (\delta, \delta) \in \mathbb{Q}_{\geq 0}^2$ be the point of intersection of $\mathcal{D}$ with $\partial \mathcal{N}(f)$. Then, $\delta_f := \delta \in \mathbb{Q}_{\geq 0}$ is the *Newton distance* of f .

For any $f(x, y) \in \mathbb{K}[x, y]$, the main face of f is either a vertex, a compact facet or a non-compact facet; the latter are a segment or a half-line, respectively. In Fig. 2, Δ_2 is the main face; in this case a segment. The Newton distance is $\frac{5}{3}$.

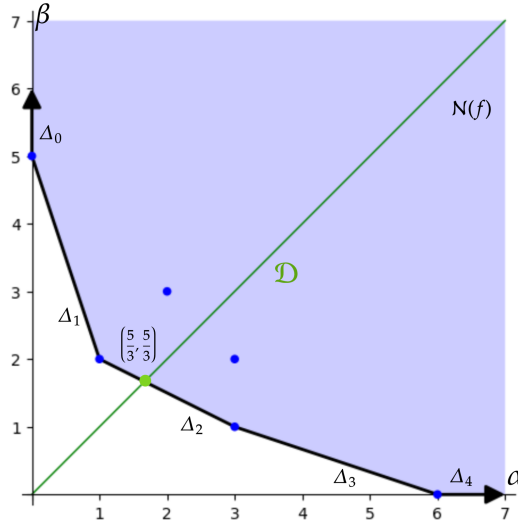


Figure 2: Newton polygon $\mathcal{N}(f)$ of $f(x, y) = y^5 + y^2x + yx^3 + x^6 + y^2x^3 + y^3x^2$, with $K = 3$ compact faces. The main face of $\mathcal{N}(f)$ is Δ_2 , and the Newton distance of f is $\delta_f = \frac{5}{3}$.

If a polynomial factors, then its Newton polygon is related to the Newton polygons of its factors, through the Minkowski sum operation. The following classical lemma illustrate this.

Lemma C.5. *Let f_1, f_2 and f be polynomials in $\mathbb{K}[x, y]$ such that $f = f_1 f_2$. Then, $\mathcal{N}(f) = \mathcal{N}(f_1) \oplus \mathcal{N}(f_2)$. Furthermore, $\mathcal{N}(f) \subseteq \mathcal{N}(f_i)$ and $\delta_{f_i} \leq \delta_f$ for $i = 1, 2$,*

Proof. This result for the *convex hull* of the Newton diagram goes back to Ostrowski [1975]. A simple proof can be found in [Gao, 2001, Lemma 2.1]. To adapt it to the *lower convex hull* case, notice that Minkowski addition is associative and commutative (from the associativity and commutativity of Euclidean vector addition). Notice also that $\mathbb{R}_{\geq 0}^2 \oplus \mathbb{R}_{\geq 0}^2 = \mathbb{R}_{\geq 0}^2$ and recall that for any sets $A \subseteq \mathbb{R}^n$ and $B \subseteq \mathbb{R}^n$, $\text{Conv}(A \oplus B) = \text{Conv}(A) \oplus \text{Conv}(B)$. Then, we have:

$$\begin{aligned}
\mathcal{N}(f) &= \text{Conv}(D(f) \oplus \mathbb{R}_{\geq 0}^2) \\
&= \text{Conv}(D(f_1 f_2) \oplus \mathbb{R}_{\geq 0}^2) \\
&= \text{Conv}(D(f_1 f_2)) \oplus \mathbb{R}_{\geq 0}^2 \\
&= (\text{Conv}(D(f_1)) \oplus \text{Conv}(D(f_2))) \oplus \mathbb{R}_{\geq 0}^2 \\
&= (\text{Conv}(D(f_1) \oplus \mathbb{R}_{\geq 0}^2) \oplus (\text{Conv}(D(f_2) \oplus \mathbb{R}_{\geq 0}^2))) \\
&= \mathcal{N}(f_1) \oplus \mathcal{N}(f_2).
\end{aligned} \tag{3}$$

For the second part of the lemma, consider a point $\mathbf{p} \in \mathcal{N}(f)$. Then, there exists $\mathbf{p}_1 \in \mathcal{N}(f_1)$ and $\mathbf{p}_2 \in \mathcal{N}(f_2)$ such that $\mathbf{p} = \mathbf{p}_1 + \mathbf{p}_2$. Since $\mathbf{p}_i \in \mathbb{R}_{\geq 0}^2$ for $i = 1, 2$, we have that $\mathbf{p} \in \mathbf{p}_i \oplus \mathbb{R}_{\geq 0}^2 \subseteq \mathcal{N}(f_i)$ for $i = 1, 2$. Since $\mathcal{N}(f) \subseteq \mathcal{N}(f_i)$, $\mathcal{D} \cap \mathcal{N}(f) \subseteq \mathcal{D} \cap \mathcal{N}(f_i)$, and hence $\delta_{f_i} \leq \delta_f$, for $i = 1, 2$. □

The following lemma relates the Newton distance of f with the Newton distance of f^m , for some positive integer m .

Lemma C.6. *Let $f(x, y) \in \mathbb{K}[x, y]$ and $m \in \mathbb{N}_{>0}$. Then, $\delta_{f^m} = m\delta_f$.*

Proof. It holds $\mathcal{N}(f^m) = \bigoplus_{i=1}^m \mathcal{N}(f) = m\mathcal{N}(f)$, where the first equality is from Lemma C.5. For the second equality, see, for example, [Schneider, 2013, Section 3.1]. Suppose that $\mathcal{D}$ intersects $\partial\mathcal{N}(f)$ at the point $\mathbf{p} = (\delta, \delta)$. Then, $\mathcal{D}$ intersects $\mathcal{N}(f^m)$ at the point $m \cdot \mathbf{p} = (m\delta, m\delta)$. Therefore, $\delta_{f^m} = m\delta_f$ as required. □

D Puiseux series

The classical theory of Newton-Puiseux expansions relates the geometry of Newton polygon to the roots of $f(x, y) \in \mathbb{K}[x, y]$, when considered as a polynomial in y (resp. x), with coefficients in $\mathbb{K}[x]$ (resp. $\mathbb{K}[y]$). Without loss of generality, we consider in the sequel that polynomials in $\mathbb{K}[x, y]$ are polynomials in y with coefficients in $\mathbb{K}[x]$. That is to say, we write $f(x, y) = \sum_{\alpha, \beta=0}^{\infty} (c_{\alpha\beta} x^\alpha) y^\beta \in (\mathbb{K}[x])[y]$. A y -root of $f(x, y)$ is an element $\phi(x)$ of the algebraic closure of $\mathbb{K}[x]$, such that $f(x, \phi(x)) = 0$. It is not sufficient for $\phi(x)$ to be an element of the field of Laurent series $\bar{\mathbb{K}}((x))$. For example, the polynomial $f(x, y) = y^2 + x \in \mathbb{K}[x, y]$ has two y -roots $\phi_{\pm}(x) = \pm\sqrt{x} = \pm x^{\frac{1}{2}}$, none of which have a Taylor expansion at the origin. Instead, y -roots are series which admit fractional exponents. The Puiseux Theorem [Casas-Alvero, 2000, Theorem 1.5.4] asserts that the algebraic closure of $\mathbb{K}((x))$ is contained in the field of Puiseux series $\bar{\mathbb{K}}\langle\langle x \rangle\rangle = \bigcup_{n \in \mathbb{N}_{>0}} \bar{\mathbb{K}}((x^{\frac{1}{n}}))$. The field of Puiseux series is the union of all Laurent series over $\bar{\mathbb{K}}$ with fractional exponents of bounded denominator.

Let $\phi(x) = \sum_{i \geq l} b_i x^{\frac{i}{n}}$ be a Puiseux series. Similarly to the case of general power series, we define the order of $\phi(x)$ to be $\text{ord}(\phi(x)) = \frac{1}{n} \min\{i \mid b_i \neq 0\}$. If $\phi(x) \in \bar{\mathbb{K}}\langle\langle x \rangle\rangle$ has finitely many terms, we define the degree of $\phi(x)$ to be $\deg \phi(x) = \frac{1}{n} \max\{i \mid b_i \neq 0\}$. By convention, we define the order of the zero series to be $\text{ord } 0 = +\infty$. Equipped with the order, the field of Puiseux series is a *valuation ring*.

If $\phi(x)$ is a y -root of $f(x, y)$ of strictly positive order, then we can compute its terms inductively (up to an arbitrary number of terms) by exploiting the geometry of the Newton polygon, following

the Newton-Puiseux algorithm [Casas-Alvero, 2000, Section 1.4]. The Newton polygon $\mathcal{N}(f)$ characterizes the first order terms of the y -roots of strictly positive order of $f(x, y)$ in the following way. Suppose that $\mathcal{N}(f)$ has K compact faces. Let Δ_i be a face of $\mathcal{N}(f)$ of weight $wt_i = (p_i, q_i)$ and let a_{ij} be a root of $F_{\Delta_i}(z)$ of multiplicity m_{ij} (see Definition C.2). Let ζ_{q_i} be a q_i -th root of unity. Then, for all $(i, j, k) \in [K] \times [r_i] \times [q_i]$, there exist m_{ij} y -roots $\phi(x)$ of strictly positive order, such that $\phi(x) = (\zeta_{q_i})^k a_{ij} x^{\frac{p_i}{q_i}} + \dots$. Moreover, there exist $m_i = h_i = q_i \sum_{j=1}^{r_i} m_{ij} = q_i l_i$ y -roots $\phi(x)$ of strictly positive order, such that $\text{ord}(\phi) = \frac{p_i}{q_i}$. In total, there are $m = \sum_{i=1}^K m_i = \sum_{i=1}^K h_i = h_f$ y -roots of strictly positive order.

The largest denominator n of the exponents of $\phi(x)$ is called the ramification index of $\phi(x)$. Let ζ_n be an n -th root of unity. Then, the map $\sigma_{\zeta_n} : \mathbb{K}((x^{\frac{1}{n}})) \rightarrow \mathbb{K}((x^{\frac{1}{n}}))$, $x^{\frac{1}{n}} \mapsto \zeta_n x^{\frac{1}{n}}$ is an automorphism of $\mathbb{K}((x^{\frac{1}{n}}))$ over $\mathbb{K}((x))$. Therefore, any Puiseux series $\phi(x)$ of ramification index n has n conjugates of the form $\sigma_{(\zeta_n)^j}(\phi(x)) = \sum_{i \geq l} b_i (\zeta_n)^{ij} x^{\frac{i}{n}}$ for $j \in [n]$.

A consequence of the Puiseux Theorem is the unique factorization of a polynomial $f(x, y)$ over $\mathbb{K}((x))$ and $\mathbb{K}(\langle x \rangle)$ [Casas-Alvero, 2000, Corollaries 1.5.5 and 1.5.6]. For a set of conjugate y -roots $\{\phi_{ij}(x)\}_{i=1}^{n_j}$ of ramification index n_j , we let $f_j(x, y) = \prod_{i=1}^{n_j} (y - \phi_{ij}(x)) \in \mathbb{K}((x))[y]$. Note that $\deg_y f_j = n_j$. Let $f(x, y)$ be a polynomial of y -degree d . Let $n_1, \dots, n_l$ be the set of ramification indices of the conjugacy classes of its y -roots. Then, there exists a unit $u \in \mathbb{K}((x, y))$, such that $f(x, y)$ factors as:

$$f(x, y) = u x^r \prod_{j=1}^l f_j(x, y) = u x^r \prod_{i=1}^d (y - \phi_i(x)),$$

where the first equality corresponds to irreducible elements of $\mathbb{K}((x))[y]$, and the second to the linear elements of $\mathbb{K}(\langle x \rangle)[y]$. In addition, the sum of the ramification indices of the y -roots of f equals the y -degree of f , that is to say, $n_1 + \dots + n_l = d_y$.

D.1 Truncated Puiseux series

For our purposes, we need to introduce a truncation of Puiseux series, which plays a central role in the analysis of Puiseux expansions of polynomials [Walsh, 2000, Poteaux and Weimann, 2021]. We use notation from Poteaux and Weimann [2021].

Definition D.1 (Truncation of Puiseux series). Let $\phi(x) = \sum_{i=k}^{\infty} b_i x^{\frac{i}{n}} \in \mathbb{K}(\langle x \rangle)$ be a Puiseux series and let $\tau \geq k$. Then, the *truncation* of $\phi(x)$ at $\tau \neq \infty$ is $\lceil \phi(x) \rceil^\tau := \sum_{i \leq \tau} b_i x^{\frac{i}{n}}$. If $\tau = \infty$, then $\lceil \phi(x) \rceil^\infty = \phi(x)$.

The singular part of a y -root $\phi(x)$ of a polynomial $f(x, y)$ is a truncation that contains the necessary information to characterize the singular locus at the origin. The computation of singular parts of the y -roots and its complexity have been discussed in several articles, e.g. Duval [1989], Walsh [2000], Poteaux and Weimann [2021].

Definition D.2 (Generalized multiplicity [Phong et al., 1999, Proof of Theorem 5]). Let $f(x, y) \in \mathbb{K}[x, y]$ be a polynomial of degree $\deg_y f = d$. Let $\phi(x) \in \mathbb{K}(\langle x \rangle)$. Let $R_f = \{\phi_i(x)\}_{i=1}^d$ be the set of all (not necessarily distinct) y -roots of f , counted with multiplicity. We say that $\phi(x)$ has generalized multiplicity $e \leq d$ with respect to R_f , if there exists exactly e y -roots $\{\phi_{i_1}, \dots, \phi_{i_e}\} \subseteq R_f$ such that $\phi(x) = \lceil \phi_{i_1}(x) \rceil^\tau = \dots = \lceil \phi_{i_e}(x) \rceil^\tau$ for some $\tau \in \mathbb{Z} \cup \{\infty\}$.

Note that if $\tau = \infty$ then, $f(x, y)$ has a y -root $\phi(x)$ of multiplicity $m_\phi = e$.

Definition D.3 (Singular part of a y -root [Walsh, 2000, Section 2]). Let $f(x, y) \in \mathbb{K}[x, y]$. Let $\phi(x) \in R_f$ be a y -root of $f(x, y)$. Then, the *singular part* $S_\phi(x)$ of $\phi(x)$ is the truncation

$$S_\phi(x) = \lceil \phi(x) \rceil^{\tau^*},$$

where τ^* is such that

$$\tau^* = \max\{\tau \in \mathbb{Z} \mid \exists \tilde{\phi}(x) \neq \phi(x) \in R_f \text{ s.t. } \lceil \tilde{\phi}(x) \rceil^\tau = \lceil \phi(x) \rceil^\tau\}.$$

We call τ^* the regularity index of $\phi(x)$.

In other words, the regularity index τ^* is the smallest truncation order which distinguishes $\phi(x)$ from the other distinct y -roots of $f(x, y)$.

Example D.4. Singular parts of y -roots can have fractional exponents. As an example, consider the Puiseux series $\phi(x) = x + x^{\frac{3}{2}} + x^{\frac{7}{4}}$. The 4-th roots of unity are $\zeta^0 = 1, \zeta^1 = i, \zeta^2 = -1$ and $\zeta^3 = -i$. We let $\phi_i = \sigma_{\zeta^i}(\phi(x))$. Then, $f(x, y) = \prod_{i=0}^3 (y - \phi_i(x))$ is a polynomial in $\mathbb{K}[x, y]$. The singular parts of this polynomial are $S_{\phi_0}(x) = S_{\phi_3}(x) = x + x^{\frac{3}{2}}$ and $S_{\phi_1}(x) = S_{\phi_2}(x) = x - x^{\frac{3}{2}}$, and their regularity indices are equal to 6.

We introduce a further subdivision of the singular part, which only retains the initial terms of the singular part with integer exponents.

Definition D.5 (Polynomial part of a y -root). Let $\phi(x)$ be a y -root of $f(x, y)$ and let S_ϕ be its singular part. The polynomial part $P_\phi(x)$ of $\phi(x)$ is the largest polynomial contained in its singular part. That is

$$P_\phi(x) = \lceil S_\phi(x) \rceil^d,$$

where $d = \max\{\tau \in \mathbb{Z} \mid \lceil S_\phi(x) \rceil^\tau \in \mathbb{K}[x, y]\}$ is the degree of the polynomial part of $P_\phi(x)$. If $d = 0$, then $P_\phi(x)$ is the zero polynomial.

Example D.6. The polynomials parts of the y -roots of $f(x, y)$ in Example D.4 are $P_{\phi_i}(x) = x$ for $0 \leq i \leq 3$.

E Deferred Proofs of Subsection 2.1

E.1 Proof of Proposition 2.6

Proof. We follow the proof of Collins [2018] and the proof of Phong et al. [1999]. Let $f(x, y)$ be an analytic function. Suppose that Δ_i is to the *right* of the main face. Let a_{ij} be a root of $F_{\Delta_i}(z)$ of multiplicity m_{ij} . Then, by Section D, we have that $\delta_f \geq h_i \geq m_{ij}$. Now suppose that $\phi(x) = a_{ij}x^{\frac{p_i}{q_i}} + \dots$ is a y -root of $f(x, y)$, such that a_{ij} is a root of multiplicity m_{ij} of $F_{\Delta_i}(z)$, and that p_i and q_i are coprime integers. Let ζ_{q_i} be a q_i -th root of unity. Then, all the q_i conjugates $\sigma_{\zeta_{q_i}^0}(\phi(x)), \dots, \sigma_{\zeta_{q_i}^{q_i-1}}(\phi(x))$ are also y -roots of $f(x, y)$. Therefore, $m_{ij} \leq \frac{h_i}{q_i}$. Furthermore, $\frac{h_i}{w_i} = \frac{q_i}{p_i}$. This implies that $h_i \geq q_i m_{ij}$ and $w_i \geq p_i m_{ij}$. Now if Δ_i is to the left of the main face, we have that $\delta_f \geq w_i \geq m_{ij}$. Therefore, facets to the left and to the right of the main face satisfy the normalization condition.

If the main face is a vertex, we are done - since all facets satisfy the normalization condition. Moreover, if the main face is a non-compact facet Δ , with facet polynomial $F_\Delta(z) = z^m$, for some $m \in \mathbb{N}_{>0}$. But $\delta_f = m$, and so Δ is normalized. Suppose now that the main face is a compact facet. Let Δ_i be this facet and let $s_i = -\frac{q_i}{p_i}$ be its slope. Then, the equation of the line cutting out Δ_i is given by

$$L(\alpha, \beta) = \frac{q_i \alpha + p_i \beta}{p_i + q_i} - \delta_f = 0.$$

Note that $L(0, h_i) = \frac{h_i p_i}{p_i + q_i} - \delta_f \leq \delta_f (\frac{p_i}{p_i + q_i} - 1) \leq 0$. Hence,

$$\delta_f \geq \frac{h_i p_i}{p_i + q_i} \geq m_{ij} \frac{p_i q_i}{p_i + q_i}. \quad (4)$$

If $m_{ij} > \delta_f$ then, $p_i + q_i > p_i q_i$. If $p_i > 1$ and $q_i > 1$, then $p_i + q_i < 2 \max(p_i, q_i) \leq p_i q_i$. Therefore, either p_i or q_i is equal to 1. Without loss of generality, we assume that $q_i = 1$. Recalling from Section D that m_{ij} corresponds to the number of y -roots of $f(x, y)$ with a given initial term $a_{ij}x^{\frac{p_i}{q_i}}$ and since $q_i = 1$, we have that $f(x, y)$ has $m_{ij} > \delta_f$ y -roots of the form $\phi(x) = a_{ij}x^{p_i} + \dots$

Let $P_f(x) \in \mathbb{K}[[x]]$ be a power series such that $P_f(x)$ has initial term $a_{ij}x^{p_i}$. Let $P_f(x)$ of maximal degree (possibly infinite) such that the generalized multiplicity of $P_f(x)$ is $e > \delta_f$. Then, consider the right equivalence $\Phi : (x, y) \mapsto (x, y - P_f(x))$. Let $\tilde{f} = f \circ \Phi$.

Then,

$$\partial \mathcal{N}(f) \cap \partial \mathcal{N}(\tilde{f}) = \{(\alpha, \beta) \mid (\alpha, \beta) \in \partial \mathcal{N}(f) \text{ s.t. } \beta \geq e\}.$$

If $P_f(x)$ lies in the set $\mathbb{K}[[x]] \setminus \mathbb{K}[x]$ then, $\phi(x) = 0$ is a y -root of multiplicity e of $\tilde{f}$. Therefore, $\delta_{\tilde{f}} = e$, since $\mathcal{D}$ intersects a non-compact horizontal facet of $\partial\mathcal{N}(\tilde{f})$.

Otherwise, $P_f(x)$ is polynomial, and the main facet $\tilde{\Delta}_i$ of $\tilde{f}$ has a finite slope $\tilde{s}_i < 0$. If $\tilde{s}_i = -\frac{\tilde{q}_i}{\tilde{p}_i}$ with $\tilde{p}_i \geq 2$ and $\tilde{q}_i \geq 2$ then, $\tilde{F}_{\tilde{\Delta}_i}$ has no root of multiplicity $\tilde{m}_{ij}$ greater than $\delta_{\tilde{f}}$. Otherwise, if, without loss of generality, $\tilde{s}_i = -\frac{1}{\tilde{p}_i}$ and $\tilde{f}_{\tilde{\Delta}_i}$ has a factor of multiplicity greater than $\delta_{\tilde{f}}$, the maximality assumption on the degree of $P_f(x)$ is contradicted. Therefore, $\tilde{f}$ is normalized, and $\tilde{f} = f \circ \Phi$ is right equivalent to f .

□

Corollary E.1. *Let $f(x, y) \in \mathbb{K}[x, y]$ and let $\phi(x) \in \mathbb{K}[[x]] \setminus \mathbb{K}[x]$ be a y -root of $f(x, y)$ of multiplicity $m > \delta_f$. Then, $\text{RLCT}_0(f) = \frac{1}{m}$.*

Proof. This follows from the last part of the proof of Proposition 2.6. □

F Deferred Proofs of Main Results

F.1 Proof of Proposition 2.7

First, we show two auxiliary lemmas used in the proof of Proposition 2.7.

Lemma F.1. *Suppose that $f(x, y) \in \mathbb{K}\{x, y\}$ is an unnormalized analytic function. Let Δ_i be its main face and m_{ij} be the multiplicity of some root of the univariate facet polynomial $F_{\Delta_i}(z)$, such that $m_{ij} > \delta_f$. Let $m_i = \deg F_{\Delta_i}(z)$. Then, $m_{ij} > \frac{m_i}{2}$.*

Proof. Let $f(x, y) \in \mathbb{K}\{x, y\}$ be any unnormalized analytic function. Then, $\mathcal{N}(f)$ has a main face Δ_i of slope $s_i = -\frac{1}{p_i}$, such that the facet polynomial $F_{\Delta_i}(z)$ has a root of multiplicity $m_{ij} > \delta_f$. We have seen in the course of the proof of Proposition 2.6, Equation 4, that $\delta_f \geq \frac{p_i m_i}{1+p_i}$. Since $p_i \geq 2$, we have that $m_{ij} > \delta_f \geq \frac{m_i}{2}$ which implies that $m_{ij} > \frac{m_i}{2}$. □

Lemma F.2. *If $f(x, y) \in \mathbb{Q}\{x, y\}$, then the normalizing power series $P_f(x)$ of $f(x, y)$ lies in $\mathbb{Q}[[x]]$.*

Proof. Suppose that $f(x, y) \in \mathbb{Q}\{x, y\}$ and is unnormalized. Suppose that $P_f(x) = ax^p$. Then, the main face of $\mathcal{N}(f)$ being Δ_i , of weight $wt_i = (p_i, 1)$, we have $p = p_i$ and $a = a_{ij}$, a root of the facet polynomial $F_{\Delta_i}(z)$. Then, the generalized multiplicity of $P_f(x)$ is $e = m_{ij}$, the multiplicity of a_{ij} as a root of $F_{\Delta_i}(z) \in \mathbb{Q}[z]$. Suppose that a_{ij} is algebraic over $\mathbb{Q}$. Then, there exists at least one algebraic conjugate a_{ij}' of a_{ij} , such that a_{ij}' is a root of $F_{\Delta_i}(z)$. Moreover, $m_{ij}' = m_{ij}$. But, by Lemma F.1, we have that $m_{ij} + m_{ij}' > m_i$, which is a contradiction. Therefore, $a = a_{ij}$ is not algebraic over $\mathbb{Q}$ and $P_f(x) \in \mathbb{Q}[[x]]$. The statement follows by induction on the number of terms of $P_f(x)$. □

Now we proceed to the proof of Proposition 2.7.

Proof. Suppose that $f(x, y) \in \mathbb{K}\{x, y\}$ is an unnormalized analytic function, and consider $P_f(x) \in \mathbb{K}\{x\}$ such that $\tilde{f}(x, y) = f(x, y - P_f(x))$ is normalized. Suppose that there exists a power series $P_f'(x) \neq P_f(x)$ whose generalized multiplicity e' is greater than or equal to the generalized multiplicity e of $P_f(x)$ and whose degree is equal to that of $P_f(x)$. Then, there exists a truncation $P_f''(x)$ of $P_f(x)$ and $P_f'(x)$ such that $\tilde{f}(x, y) := f(x, y - P_f''(x))$ is an unnormalized power series, with main face $\tilde{\Delta}_i$. Let $\tilde{F}_{\tilde{\Delta}_i}(z)$ be the facet polynomial associated to $\tilde{\Delta}_i$. Let $\tilde{m}_i$ be its degree, and let $\delta_{\tilde{f}}$ be the Newton distance of $\mathcal{N}(f)$. Then $\tilde{F}_{\tilde{\Delta}_i}(z)$ has two roots of multiplicity e and e' , such that $e' \geq e > \delta_{\tilde{f}}$. By definition, $e + e' \leq \tilde{m}_i$. But by Lemma F.1, $e + e' > \frac{\tilde{m}_i}{2} + \frac{\tilde{m}_i}{2} = \tilde{m}_i$, which is a contradiction. Hence, $P_f(x)$ must be unique. Rationality of $P_f(x)$ follows from Lemma F.2. □

Remark F.3. By Lemma F.2, if we seek to compute the local RLCT of a polynomial $f(x, y)$ with rational coefficients, we do not need to work over algebraic field extensions of $\mathbb{Q}$. At each iteration of the algorithm, we only need to care about the unique (by Proposition 2.7) rational root of $F_{\Delta_i}(z)$ of multiplicity $m_{ij} > \delta_f$. This means that we do not need a full factorization of $F_{\Delta_i}(z)$ at each step, merely a square-free factorization.

F.2 Proof of Theorem 2.11

Since $P_f(x) \in \mathbb{Q}[[x]]$, it follows that $P_f^{\text{fin}}(x) \in \mathbb{Q}[x]$. The following Lemma explains how to compute the local RLCT at the origin of $f(x, y)$ using a right equivalence defined in terms of $P_f^{\text{fin}}(x)$.

Proof. Let $f(x, y) \in \mathbb{Q}[x, y]$, with normalizing power series $P_f(x) \in \mathbb{Q}[x]$. Let $P_f^{\text{fin}}(x)$ be as above, and construct $\tilde{f}(x, y) = f(x, y - P_f^{\text{fin}}(x))$. If $\tilde{f}$ is normalized, then $P_{\tilde{f}}^{\text{fin}}(x) = P_f(x)$, by Proposition 2.7. Then, by Proposition 2.5, we have that $\text{RLCT}_0(f) = \frac{1}{\delta_{\tilde{f}}}$. Otherwise, $\tilde{f}$ is not normalized. Then, it must be that $P_f(x)$ is a y -root of $f(x, y)$ and $P_f^{\text{fin}}(x)$ is its singular part. Let Δ be the main face of $\mathcal{N}(\tilde{f})$. Let $\tilde{s} = \frac{-1}{p}$ be its slope. Let $a \in \mathbb{Q}$ be the root of $\tilde{F}_{\Delta}$, such that the multiplicity m of a is maximal, and $m > \delta_f$. By the maximality assumption on the degree of singular parts (see Definition D.3), there exists a y -root $\tilde{\phi}(x)$ of $\tilde{f}$ of multiplicity m , such that the initial term of $\tilde{\phi}(x)$ is ax^p . Then $P_f(x) = P_f^{\text{fin}}(x) + \tilde{\phi}(x)$ is the normalizing power series of $f(x, y)$ and it corresponds to a y -root of $f(x, y)$ of order m . Therefore, $\text{RLCT}_0(f) = \frac{1}{m}$. $\square$

F.3 Proof of Theorem 2.12

An upper bound on the degree d of the finite part $P_f^{\text{fin}}(x)$ of the normalizing power series $P_f(x)$ of $f(x, y)$, is an upper bound on the number of iterations of Algorithm 1. Consider two y -roots $\phi_1(x) = P_f^{\text{fin}}(x) + \tilde{\phi}_1(x)$ and $\phi_2(x) = P_f^{\text{fin}}(x) + \tilde{\phi}_2(x)$, such that $\tilde{\phi}_1(x)$ and $\tilde{\phi}_2(x)$ are not necessarily distinct Puiseux series and $\text{ord } \tilde{\phi}_i(x) > d$, for $i = 1, 2$. Then, $\text{ord}(\phi_1(x) - \phi_2(x)) = \min\{\text{ord } \tilde{\phi}_1(x), \text{ord } \tilde{\phi}_2(x)\} > d$. Therefore, finding an upper bound on the order of the differences of y -roots of $f(x, y)$ provides an upper bound on d . A natural approach is to consider the discriminant in y of $f(x, y)$.

Before deriving upper bounds on the complexity of our algorithms, we remind the reader of some basic notions. We recall the definitions and basic properties of resultants and discriminants, as well as the definition of the square-free factorization of a polynomial. A classical reference for the former can be found in Cox et al. [1997], Chapter 3.

Definition F.4 (Sylvester matrix, [Cox et al., 1997, Chapter 3, Section 6, Definition 2]). Let R be an integral domain. Let $f(z) = a_m z^m + \dots + a_1 z + a_0$ and $g(z) = b_n z^n + \dots + b_1 z + b_0$ be two polynomials in $R[z]$ of degree m and n respectively, such that $a_m \neq 0$ and $b_n \neq 0$. The Sylvester matrix $\text{Syl}(f, g)$ of f and g is the $(m+n) \times (m+n)$ matrix defined as

$$\text{Syl}(f, g) = \begin{pmatrix} a_0 & a_1 & \cdots & a_{m-1} & a_m & 0 & \cdots & 0 \\ 0 & a_0 & a_1 & \cdots & a_{m-1} & a_m & \cdots & 0 \\ \vdots & & \ddots & \ddots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_0 & a_1 & \cdots & a_{m-1} & a_m \\ b_0 & b_1 & \cdots & b_{n-1} & b_n & 0 & \cdots & 0 \\ 0 & b_0 & b_1 & \cdots & b_{n-1} & b_n & \cdots & 0 \\ \vdots & & \ddots & \ddots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & b_0 & b_1 & \cdots & b_{n-1} & b_n \end{pmatrix} \in R^{(m+n) \times (m+n)}.$$

Definition F.5 (Resultant, [Cox et al., 1997, Chapter 3, Section 6, Definition 2]). The resultant $\text{res}(f, g)$ of f and g is the determinant of their Sylvester matrix:

$$\text{res}(f, g) = \det(\text{Syl}(f, g)).$$

Proposition F.6. Let $f(z)$ and $g(z)$ be as above. Then, the resultant of f and g can be expressed as:

$$\text{res}(f, g) = a_m^n b_n^m \prod_{i,j} (x_i - y_j)$$

where $x_1, \dots, x_m$ and $y_1, \dots, y_n$ are the roots of f and g respectively, counted with multiplicity, lying in the algebraic closure $\bar{F}$ of $F = \text{Frac}(R)$.

Proof. See [Cox et al., 1998, Chapter 3, Section 1]. □

Note that, by Proposition F.6, f and g have a common root if and only if $\text{res}(f, g) = 0$.

Definition F.7 (Discriminant [Cox et al., 1998, Chapter 3, Section 6, Exercise 16]). The discriminant $\text{disc}(f)$ is defined as

$$\text{disc}(f) = (-1)^{\frac{n(n-1)}{2}} \frac{1}{a_n} \text{res}(f, f') = (-1)^{\frac{n(n-1)}{2}} a_n^{2n-2} \prod_{i < j} (x_i - x_j)^2,$$

where $x_1, \dots, x_n$ are roots of $f(z)$.

Note that f has a multiple root if and only if $\text{disc}(f) = 0$.

We give an upper bound on the number of iterations of Algorithm 1 given a polynomial $f(x, y) \in \mathbb{Q}[x, y]$, by bounding the degree d of $P_f^{\text{fin}}(x) \in \mathbb{Q}[x]$. In the following, we denote by $|\cdot| = e^{-\text{ord}(\cdot)} : \mathbb{C}\langle\langle x \rangle\rangle \mapsto \mathbb{R}_{>0}$ the absolute value induced by the valuation $\text{ord}(\cdot)$ on $\mathbb{C}\langle\langle x \rangle\rangle$.

Proposition F.8 (Root separation for rational bivariate polynomials). *Let $f(x, y) = a_{d_y}(x)y^{d_y} + \dots + a_1(x)y + a_0(x) \in \mathbb{Q}[x, y]$ be a square-free polynomial, such that $\deg_y f = d_y$ and $\deg_x f = d_x$. Let $0 < |\phi_1| \leq |\phi_2| \leq \dots \leq |\phi_{d_y}|$ be the y -roots of $f(x, y)$. Let Ω be any set of k pair of indices (i, j) , such that $1 \leq i < j \leq d_y$. Then,*

$$\sum_{(i,j) \in \Omega} \text{ord}(\phi_i - \phi_j) \leq (d_y + k - \frac{1}{2})d_x.$$

Proof. We adapt the proof of [Tsigaridas and Emiris, 2008, Theorem 7], from univariate polynomial with integer coefficients to bivariate polynomials with rational coefficients. Consider the multiset $\bar{\Omega} = \{j \mid (i, j) \in \Omega\}$ of cardinality $|\bar{\Omega}| = k$. We denote the order of the leading term of $f(x, y)$ by $h = \text{ord}(a_{d_y}(x))$. We begin by observing that, for any $i \in [d_y]$,

$$-h \leq \text{ord}(\phi_i) \leq d_x \iff e^{-d_x} \leq |\phi_i| \leq e^h. \quad (5)$$

Next, we obtain a bound on the absolute value of the product of the roots of $f(x, y)$. Note that

$$|\phi_i| \geq 1 \iff \text{ord} \phi_i \leq 0.$$

Let $d_y^- \leq d_y$ be the number of y -roots of order ≤ 0 , and denote by $\{\phi_i^-\}_{i=1}^{d_y^-}$ the roots of negative order. We have the following relation between h and the y -roots of negative order,

$$-h = \sum_{i=1}^{d_y^-} \text{ord} \phi_i^- \iff e^h = \exp(-\sum_{i=1}^{d_y^-} \text{ord} \phi_i^-) = \prod_{i=1}^{d_y^-} |\phi_i^-|.$$

Thus,

$$\prod_{i=1}^{d_y} |\phi_i| \leq \prod_{i=1}^{d_y} \max\{|\phi_i|, 1\} = \prod_{i=1}^{d_y^-} |\phi_i^-| = e^h. \quad (6)$$

Furthermore, since the absolute value $|\cdot|$ is induced by a valuation, it satisfies the ultrametric inequality:

$$\forall \phi, \phi' \in \mathbb{C}\langle\langle x \rangle\rangle, |\phi - \phi'| \leq \max\{|\phi|, |\phi'|\}. \quad (7)$$

Recall, by Definition F.7, that $\text{disc } f = (-1)^{\frac{d_y(d_y-1)}{2}} a_{d_y}(x)^{2d_y-2} \prod_{i < j} (\phi_i - \phi_j)^2$. Therefore,

$$\begin{aligned} |\text{disc } f| &= (e^{-h})^{2d_y-2} \prod_{i < j} |\phi_i - \phi_j|^2 \\ &= (e^{-h})^{2d_y-2} \prod_{(i,j) \in \Omega} |\phi_i - \phi_j|^2 \prod_{(i,j) \notin \Omega} |\phi_i - \phi_j|^2. \end{aligned}$$

We consider the product $\prod_{(i,j) \notin \Omega} |\phi_i - \phi_j|$ and apply $\binom{d_y}{2} - k$ times the ultrametric inequality 7. Thus,

$$\begin{aligned} \prod_{(i,j) \notin \Omega} |\phi_i - \phi_j| &\leq |\phi_1|^0 |\phi_2|^1 \cdots |\phi_{d_y}|^{d_y-1} \left(\prod_{j \in \Omega} |\phi_j| \right)^{-1} \\ &\leq \left(\prod_{i=1}^{d_y^-} |\phi_i^-| \right)^{d_y-1} \left(\prod_{j \in \Omega} |\phi_j| \right)^{-1} \\ &\leq e^{h(d_y-1)} e^{kd_x}, \end{aligned}$$

where we used Inequalities 6 and 5. Then,

$$\begin{aligned} \prod_{(i,j) \in \Omega} |\phi_i - \phi_j| &= e^{h(d_y-1)} \left(\prod_{(i,j) \notin \Omega} |\phi_i - \phi_j| \right)^{-1} \sqrt{|\text{disc}(f)|} \\ &\geq e^{-kd_x} \sqrt{|\text{disc}(f)|} \end{aligned}$$

Now, by expressing $\text{disc } f$ in terms of the determinant of the $(2d_y - 1) \times (2d_y - 1)$ matrix $\text{Syl}(f, f')$, we obtain the following upper bound on the order of $\text{disc } f$:

$$\text{ord}(\text{disc } f) \leq \deg_x(\text{disc } f) \leq (2d_y - 1)d_x.$$

Altogether,

$$\prod_{(i,j) \in \Omega} |\phi_i - \phi_j| \geq e^{-kd_x} e^{-(d_y - \frac{1}{2})d_x} = e^{-(d_y + k - \frac{1}{2})d_x}.$$

Finally, we get:

$$\sum_{(i,j) \in \Omega} \text{ord}(\phi_i - \phi_j) \leq (d_y + k - \frac{1}{2})d_x.$$

□

Now we can prove Theorem 2.12.

Proof. Given a polynomial f , consider its square-free factorization, $\tilde{f}$. This is a polynomial of degree $\deg_y \tilde{f} \leq d_y$ and $\deg_x \tilde{f} \leq d_x$. Since taking the square-free part of f does not affect the finite part of its normalizing power series, $P_{\tilde{f}}^{\text{fin}}(x) = P_f^{\text{fin}}(x)$. If $\tilde{f}$ has a single y -root of strictly positive order, then $P_{\tilde{f}}^{\text{fin}}(x) = 0$, therefore $\deg P_{\tilde{f}}^{\text{fin}}(x) = -\infty$ and the corollary follows trivially. Now suppose that $P_{\tilde{f}}^{\text{fin}}(x)$ has generalized multiplicity $e \geq 2$. Let $d = \deg P_{\tilde{f}}^{\text{fin}}(x)$. Then, there exist $\phi_{i_1}(x), \dots, \phi_{i_e}(x)$ y -roots such that $[\phi_{i_j}(x)]^d = [\phi_{i_{j'}}(x)]^d$, for all $(j, j') \in [e]^2$. Since $\tilde{f}$ is square-free, at least 2 of the e y -roots are distinct. Let $\phi_{i_j}(x)$ and $\phi_{i_{j'}}(x)$ be such roots. Then, $(d_y + \frac{1}{2})d_x \geq \text{ord}(\phi_{i_j} - \phi_{i_{j'}}) > d$, by Proposition F.8. □

G Deferred Proofs of Algorithmic Results

Consider the while loop defined by lines 5–14 of Algorithm 1. By construction, this loop runs for at most $B = (d_y + \frac{1}{2})d_x$. Fix an input $f \in \mathbb{Q}[x, y]$. Let $I \leq B$ be the total number of iterations ran by the algorithm before exiting the while loop, for input $f(x, y)$. Let $f^{(0)}(x, y) = f(x, y)$. Let $(b^{(0)}, p^{(0)}) = (0, 0)$. For $i \in [I]$, let $(b^{(i)}, p^{(i)}, m^{(i)})$ be the triplet computed in line 10. Let $f^{(i+1)}(x, y) = f^{(i)}(x, y - b^{(i)}x^{p^{(i)}})$ be the change of variables in line 11, and let $B^{(i+1)} = B^{(i)} - (p^{(i)} - p^{(i-1)})$. Observe that $B^{(i)} = B - p^{(i)}$. Finally, let $P^{(i)}(x) = \sum_{j=0}^i b^{(j)}x^{p^{(j)}}$. Furthermore, denote by $\Delta^{(i)}$ the main face of $\mathcal{N}(f^{(i)})$.

Lemma G.1. *Let $f(x, y) \in \mathbb{Q}[x, y]$ be an input of Algorithm 1. $P_f(x) \in \mathbb{Q}[[x]] \setminus \mathbb{Q}[x]$ if and only if there exists $N \in \mathbb{N}_{>0}$, such that $f^{(N)}$ is not normalized and $B^{(N)} \leq 0$.*

Proof. Let $f(x, y) \in \mathbb{Q}[x, y]$ be an un-normalized polynomial, such that $\deg_y f = d_y$ and $\deg_x f = d_x$. Let $B = (d_y + \frac{1}{2})d_x$. Suppose that such an N exists. Let $\phi_1(x), \dots, \phi_n(x)$ be the y -roots of strictly positive orders, whose initial coefficients are roots of $F_{\Delta^{(0)}}$. Let $m_1 \geq \dots \geq m_n$ be their multiplicities as Puiseux y -roots of $f(x, y)$. Then $n \leq \deg_z F_{\Delta^{(0)}} = \sum_{i=1}^n m_i := m$. Now let $m^{(0)} = m_1^{(0)} \geq \dots \geq m_{r^{(0)}}^{(0)}$, for $r^{(0)} \leq n$ be the multiplicities of the distinct roots of $F_{\Delta^{(0)}}(z)$. Likewise, $m = \sum_{j=1}^{r^{(0)}} m_j^{(0)}$. Now, since $B^{(N)} \leq 0$, it must be that $\deg P^{(N)}(x) = p^{(N)} \geq B$. By Proposition F.8, there exist no two distinct y -roots $\phi_i(x)$ and $\phi_j(x)$ as above, such that $[\phi_i(x)]^{p^{(N)}} = [\phi_j(x)]^{p^{(N)}}$. In other words, all the y -roots are separated, and $F_{\Delta^{(N)}}(z)$ has $r^{(N)} = n$ distinct roots, each with multiplicity $m_i^{(N)} = m_i$, for $i \in [n]$. Since $f^{(N)}$ is not yet normalized, we have that $m_1 > \delta_{f^{(N)}}$. So $\phi_1(x) \in \mathbb{Q}(\langle x \rangle)$ is a y -root of $f(x, y)$ of multiplicity $m_1 > \delta_f$. Now, Lemma F.1 implies that $m_1 > m_i$ for all $1 < i \leq n$. Therefore, let $g(x, y)$ be the minimal polynomial of $\phi_1(x)$. Then, $f(x, y) = g(x, y)^{m_1} h(x, y)$, for some polynomial $h(x, y) \in \mathbb{Q}[x, y]$. By Lemma C.5 and Lemma C.6, we have that $\delta_g < 1$. With this geometric constraint, $\mathcal{N}(g)$ must be spanned by a single facet $\Delta = (\mathbf{p}_0, \mathbf{p}_1)$, where $\mathbf{p}_0 = (0, 1)$ and $\mathbf{p}_1 = (a, 0)$, for some $a \in \mathbb{N}$. Then, $g(x, y)$ has a single y -root of strictly positive order $\phi_1(x)$. By [Banderier and Drmota, 2013, Theorem 1], we have that $\phi_1(x) \in \mathbb{Q}[[x]] \setminus \mathbb{Q}[x]$. Moreover, the generalized multiplicity m_1 of $P_f(x)$ is greater than δ_f . Therefore $P_f(x) = \phi_1(x)$. Now suppose that $P_f(x) \in \mathbb{Q}[[x]] \setminus \mathbb{Q}[x]$. Then, for all $i \in \mathbb{N}_{>0}$, we have that $m^{(i)} > \delta_{f^{(i)}}$, and therefore $f^{(i)}$ is not normalized. $\square$

G.1 Proof of Theorem 3.1

Proof. Suppose that $P_f(x) \in \mathbb{Q}[x]$. By Lemma G.1, there exists $N \in \mathbb{N}_{>0}$ such that $f^{(N)}$ is normalized and $B^{(N)} > 0$. By Proposition 2.7, $P^{(N)}(x)$ is the normalizing power series of $f(x, y)$, and the algorithm returns the correct output. Suppose that $P_f(x) \in \mathbb{Q}[[x]] \setminus \mathbb{Q}[x]$. By Corollary E.1, $\text{RLCT}_0(f) = \frac{1}{m}$. Then, for any finite truncation $P(x)$ of $P_f(x)$, $\tilde{f}(x, y) = f(x, y - P(x))$ is not normalized. Therefore, Algorithm 1 exits the while loop defined by lines 5–14 and correctly outputs $\text{RLCT}_0(f)$. $\square$

H Auxiliary Algorithms

Here, we present the Normalized algorithm (Algorithm 2), a necessary sub-routine of Algorithm 1. This algorithm takes as input a polynomial $f(x, y) \in \mathbb{Q}[x, y]$, and returns True if the polynomial is normalized or a triple of integers (b, p, m) , where b is the root of maximal multiplicity m of the facet polynomial F_Δ of the main face Δ , and (p, q) is the weight of Δ . This triple is then used to define the change of variables in Algorithm 1.

Note that since we know that $P_f(x) \in \mathbb{Q}[[x]]$ by Lemma F.2, it is sufficient to compute a square-free factorization of $F_{\Delta_i}(z) \in \mathbb{Q}$, as opposed to a full factorization in $\bar{\mathbb{Q}}$.

Algorithm 2: Normalized

Input: A polynomial $f(x, y) \in \mathbb{Q}[x, y]$, its Newton polygon $\mathcal{N}(f)$, and its Newton distance δ_f .

Output: True if the main facet Δ of $\mathcal{N}(f)$ is normalized; otherwise a tuple $(b, p, m) \in \mathbb{Q} \times \mathbb{N} \times \mathbb{N}$.

```
1 Identify the main face  $\Delta$  of  $\mathcal{N}(f)$ 
2 Compute  $F_\Delta(z)$ 
3 if  $F_\Delta(z)$  is a single term then
4   return True
5 else
6   Compute the weight  $wt = (p, q)$  of  $\Delta$ 
7   if  $q \neq 1$  then
8     return True
9   else
10    Compute the square-free factorization  $F_\Delta(z) = \prod_{i=1}^m (F_{\Delta,i}(z))^i$ 
11    if  $m \leq \delta_f$  then
12      return True
13    else
14      Let  $F_{\Delta,m}(z) = (z - b)$ 
15      return  $(b, p, m)$ 
16    end
17  end
18 end
```
